

AlphaStream

Systematic Multi-Asset Funds with News-Sentiment Analytics

FINS3645 Financial Market Data Design & Analysis | Part B report | z5590113

Live app: <https://alphastream-z5590113projectb.streamlit.app/>

Code: https://github.com/kedar-mehta-1/z5590113_projectB

Executive summary

AlphaStream is an investment app which offers 17 systematically managed funds. These are all built from optimal portfolios from over 50 US large-cap equities and 10 cryptocurrencies alongside a standalone news-sentiment index spanning ten equity sectors. Every fund is assessed through a walk-forward, out-of-sample (OOS) backtest from 2 January 2022 to 29 December 2023. Only data available before each monthly rebalance is used to prevent lookahead bias. The best equity funds earned positive OOS returns (Equal Weight +5.97% CAGR, Sharpe 0.34; Risk Parity +3.13%, Sharpe 0.19) whilst the crypto funds lost 11% to 24% a year in the 2022-23 crypto bear market. The combined funds metrics sit between the two. Furthermore, a finance-aware extension of VADER was developed as the FinSent lexicon. This resulted in a 19% decrease in false neutral score of zero (48% to 29% neutral). However, fusing that signal into the equity funds did not add value out of sample but a sentiment tilt improved on its Maximum-Sharpe base (Sharpe -0.23 to -0.15) but remained negative. Moreover, a long-short sentiment-momentum fund lost money while its equal-weight base gained 5.97%. Beyond the required baseline, the project carries a various range of documented extensions. These include the FinSent lexicon, a 17-fund shelf with five distinct methods, an original design system, and a results-first app architecture with turnover reported as a cost proxy, with further details being provided in Section 4. The Streamlit app delivers the full investor journey through allowing a comparison of funds, the reading of fact sheets, setting allocations, and inspecting the sentiment analytics.

1. The funds and the backtest design

1.1 Product and data foundation

AlphaStream core product offering is a set of investable funds from which the app earns a 0.40% per-annum management fee charged in the backtest. The data foundation was Part A and is reused here. Namely, 50 US large-cap stocks across ten sectors, ten cryptocurrencies, and 146,836 unique daily headlines for the equities, all 2020-2023 which were all loaded through the course data-access helper and were never committed to the repository. Returns are computed within each distinct panel first. Equities on their 252-day calendar, crypto on its own daily calendar and the crypto returns are then left merged onto the equity trading calendar, so weekend-only crypto moves that an equity-day fund could not trade on are excluded by construction. The sample ends 31 December 2023 with the ten stray 2024-01-01 crypto rows being dropped and counted. Table A1 in the Appendix reproduces the integrity audit demonstrating no duplicate price rows in either panel, 1,800 missing equity ticker-days (36 business dates across all 50 ticker which are US holidays rather than data gaps), 384 outlier returns flagged and kept as real events, and 2,847 exact duplicate headlines removed.

The text panel is the second foundation. Each of the 149,683 raw headlines is date-stamped in timezone-aware UTC while the price calendar is timezone-naïve. Thus, the two are normalized to a common dtype before any alignment. Exact duplicates on ticker, date and title are removed (2,847 rows), leaving 146,836 unique headlines. News has many rows per ticker-date by design, so a ticker-date-only duplicate check would have flagged everything. Every headline is then mapped to its equity trading day or the next trading day, via a search on the trading day index. The raw headline text is preserved untouched. VADER and its FinSent extension depend on the original word order and on punctuation and stopwords, so no stopword stripping or stemming is applied. Furthermore, Part A's Station 2 features of the cross-sector news-flow spillover index (headline-count co-movement across the ten sectors, mean 0.17 and a peak of 0.71 in crisis weeks) and a crypto-to-equity lead-lag diagnostic (Section 4.4) are reused directly.

1.2 Fund lineup and optimisation methods

Each (asset family, method) pair is one fund with its own fact sheet which can be opened on the Streamlit app and is what a user invests in. The shelf holds 17 funds: five equity-only, five crypto-only, five combined, and two sentiment-fusion funds (Section 4.2). Five optimisation methods are used which include Equal Weight (EW), Minimum Variance (MV), Maximum Sharpe (MS), Risk Parity (RP) and Minimum CVaR (CV). All methods are long-only with per-family per-name caps of 10% (equity, 50 names), 20% (crypto, ten names, so at least five names are held), and 10% (combined, 60 names). The caps are per family deliberately: a flat 10% cap on ten names would collapse every crypto method to equal weight.

1.3 Backtest design and assumptions

In terms of backtest design all funds use one walk-forward scheme to eliminate look-ahead bias. This is accomplished through a decision date which is a month-end trading day with at least 504 prior trading days of history. From this the weights are solved from history strictly before that date and held until the next month. Portfolio return is then calculated as the dot product of yesterday's weights with today's asset returns.

The first live backtest date is 2 January 2022 and therefore the OOS period is exactly two years (501 trading days). Any assets with fewer than 120 valid observations are excluded from a given rebalance. Returns are also calculated as the net of the 0.40% annual fee whilst transaction costs are assumed to be zero. However, one-way average monthly turnover per fund is used as a transactions cost proxy. The distinctive choices for each parameter and setting can be seen in Table 1 below.

Beyond this, two design choices further aim to remove any leaks between the estimation window and the holding month. Firstly, the returns history used to solve a decision is indexed strictly before the decision date. The decision day's own return is not in the window, and a date becomes eligible only once 504 prior rows exist. Secondly, the sentiment signal used by the fusion funds is shifted by one trading day. As a result, a Saturday or Monday headline that is aligned to Monday only becomes usable for Tuesday's trade (Section 3.2). Furthermore, the OOS period is exactly two years, which is short enough that one regime dominates the numbers and this is specifically flagged as a limitation in Section 6.

Table 1. Walk-forward out-of-sample backtest design for all 17 AlphaStream funds.

Parameter	Setting
Data window	2 Jan 2020 to 29 Dec 2023 (equity calendar, 1,006 days)
Estimation window	504 trading days (2 years), expanding
First live (OOS) date	2 Jan 2022
OOS period	2 Jan 2022 to 29 Dec 2023 (501 trading days)
Rebalance frequency	Month-end, monthly
Information available at decision t	Data strictly before t

Covariance	Ridge-shrunk sample covariance, 0.25 shrinkage, 504-day lookback
Return forecast	Historical mean over the estimation window
Caps (per name)	10% equity / 20% crypto / 10% combined
Long-only / shorting	Long-only funds; momentum fund is dollar-neutral long-short
Fees	0.40% p.a. management fee, netted monthly
Transaction costs	Zero (turnover reported as a cost proxy)
Sharpe annualisation / rf	sqrt(252), risk-free rate = 0
Missing-name rule	Drop names with fewer than 120 valid observations

2. Out-of-sample results and fund fact sheets

2.1 Performance across funds and methods

Table 2 below reports OOS performance for all 17 funds. Out of these differing results, the standout results are the differences in performance per asset classes. The best equity funds generated positive returns of same whilst every crypto fund lost between 11% and 24% a year in the 2022-23 crypto bear market. The combined funds naturally lie between the two. A key result however is that the maximum-Sharpe method is the worst performer in every family, most likely as its mean forecast is an extremely noisy input.

Within the equity family the ranking is stable and interpretable. Equal Weight prevails because with 50 names each bet is small and forecasts can cancel out; Risk Parity and Minimum Variance are close behind, both delivering their promised risk control (13.6-16.3% volatility against 17.7% for Equal Weight). Maximum Sharpe is the only equity fund with a double-digit loss (-4.14% CAGR) and the deepest drawdown (-25.7%), which repeats again in the crypto and combined families. The crypto results are the clearest evidence that the period and not the method drove the losses. Even Minimum Variance, which is the most defensive construction, lost -11.3% a year because the whole asset class fell. The two-year OOS window is a single long crypto bear market, so the crypto shelf is a stress test more than a track record.

The combined funds expose an interesting diversification lesson. Adding a 20%-capped crypto weight to the 50-ticker equity portfolio decreased CAGR to +2.67% (Equal Weight) against the +5.97% equity-only and dragged every combined Sharpe below its equity counterpart. Correlation offered no rescue because in this window equities and crypto fell together in 2022. The honest product implication which is further discussed in Section 6 suggests that a combined portfolio needs either a much smaller crypto cap or a volatility control measure.

Turnover also substantiates earlier claims but from the cost side. Maximum-Sharpe and Minimum-CVaR funds trade 25-40% of the portfolio each month (combined Maximum Sharpe 0.31, Minimum CVaR 0.40), while Equal Weight and Risk Parity are almost buy-and-hold (0.00 and 0.02-0.05).

Table 2. Out-of-sample performance, net of the 0.40% management fee, 2 Jan 2022 to 29 Dec 2023. Volatility is annualised; Sharpe uses rf = 0; turnover is average monthly one-way portfolio turnover.

Fund	CAGR	Volatility	Sharpe	Max DD	Turnover
Equity Equal Weight	5.97%	17.71%	0.338	-20.32%	0.00
Equity Risk Parity	3.13%	16.29%	0.192	-19.25%	0.02
Equity Minimum Variance	1.96%	13.62%	0.144	-16.80%	0.12
Equity Minimum CVaR	-0.20%	14.15%	-0.014	-16.31%	0.40
Equity Maximum Sharpe	-4.14%	18.33%	-0.226	-25.74%	0.34
Crypto Minimum Variance	-11.32%	56.02%	-0.202	-64.04%	0.09
Crypto Risk Parity	-20.87%	57.62%	-0.362	-65.63%	0.01
Crypto Minimum CVaR	-20.89%	57.65%	-0.362	-64.25%	0.21

Crypto Equal Weight	-22.24%	58.53%	-0.380	-65.87%	0.00
Crypto Maximum Sharpe	-23.99%	57.77%	-0.415	-64.90%	0.25
Combined Risk Parity	2.38%	17.10%	0.139	-21.80%	0.02
Combined Equal Weight	2.67%	20.91%	0.128	-26.94%	0.00
Combined Minimum Variance	0.97%	13.83%	0.070	-17.52%	0.11
Combined Minimum CVaR	-0.01%	14.16%	-0.001	-16.31%	0.40
Combined Maximum Sharpe	-6.08%	20.07%	-0.303	-29.28%	0.31
Sentiment Equal Weight	-0.72%	7.61%	-0.094	-13.11%	1.63
Sentiment Maximum Sharpe	-2.85%	18.44%	-0.154	-24.40%	0.62

Figure 1 shows growth of \$1 for the combined family. Equal Weight and Risk Parity end the period above \$1 whilst Minimum Variance holds near \$1 with the shallowest drawdowns. Maximum Sharpe and Minimum CVaR end below it. The sharpest drop is the 2022 bear-market leg that all long-only funds suffered, and recovery is partial within the sample, which is why even the winning funds end only modestly above their starting value.



Figure 1. Growth of \$1 (net of fees) for the five combined funds, monthly rebalance, 2 Jan 2022 to 29 Dec 2023. Indexed to \$1.00 at the first live backtest date.

2.2 Drawdowns and weights over time

Figure 2 below shows the drawdown path of the combined Maximum Sharpe fund which is the riskiest out of its counterparts. Its peak-to-trough loss of 29.3% is deeper than the other combined funds (Minimum CVaR -16.3%, Minimum Variance -17.5%, Equal Weight -26.9%, Risk Parity -21.8%). The drawdown figures tell a potential investor which fund they can tolerate. Minimum CVaR and Minimum Variance are the smooth options whilst Maximum Sharpe is the aggressive one.

The weights-over-time chart in Figure 3 below explains these drawdowns mechanically. Risk Parity holds a nearly fixed sector allocation all period and its risk budget does not move when prices move. This is exactly why it produced the most stable risk-adjusted numbers. Minimum Variance rotates toward defensive sectors (Healthcare, Utilities, Consumer) during the 2022 sell-off which keeps its drawdown shallow. In contrast, Maximum Sharpe chases the sectors with the highest recent average return, but it is systematically late. This is why the momentum-style behaviour of the maximum-Sharpe fund is a cautionary result rather than an endorsement.

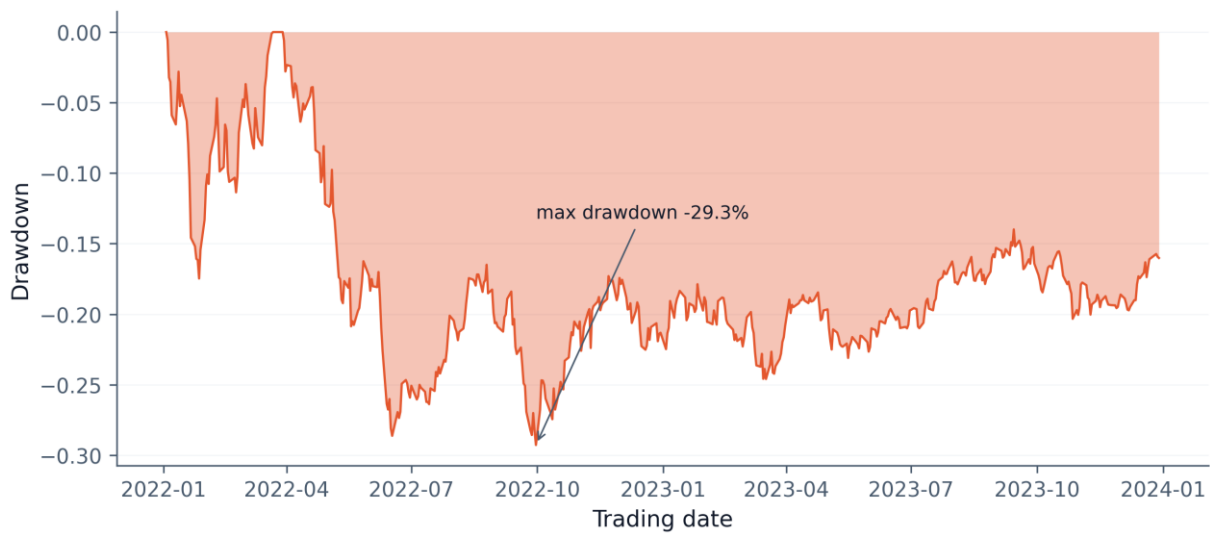


Figure 2. Drawdown (peak-to-trough loss, net of fees) of the combined Maximum Sharpe fund, 2 Jan 2022 to 29 Dec 2023.

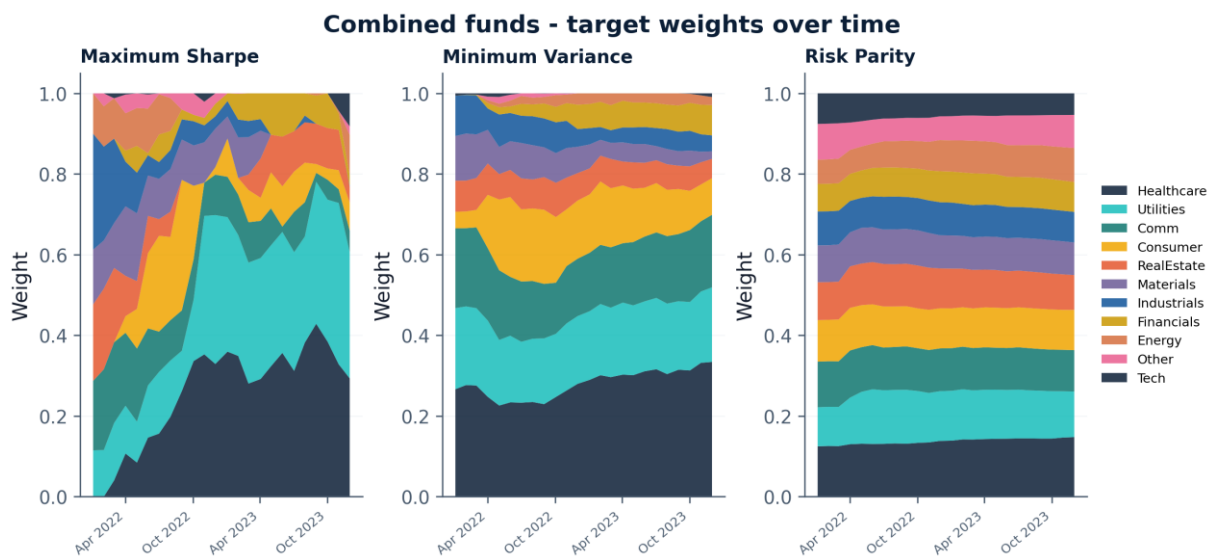


Figure 3. Sector-aggregated target weights over time for combined Maximum Sharpe, Minimum Variance and Risk Parity, monthly rebalances, 2 Jan 2022 to 29 Dec 2023.

2.3 Sharpe and risk-return comparison

Figure 4 below ranks all 17 funds by OOS Sharpe ratio. Only three combined funds are clearly above zero and every crypto fund is below -0.20. The bar chart is the honest summary that diversification across asset classes did not help in this two-year window because the crypto part experienced significant negative performance.

Figure A3 in the appendix plots the same 17 funds in return-volatility space with iso-Sharpe lines. The chart makes two points the bar plot cannot. The best risk-adjusted equity funds (Equal Weight, Risk Parity) sit at moderate risk with positive returns. In contrast, the crypto funds cluster in the bottom-right paying 56-59% volatility for returns between -11% and -24% significantly below the iso-Sharpe line of zero. This serves to be strong visual argument against a large crypto allocation in this period. Importantly, the sentiment-momentum fund is the outlier in the bottom-left as it has low volatility (7.6%) and only slightly negative returns.

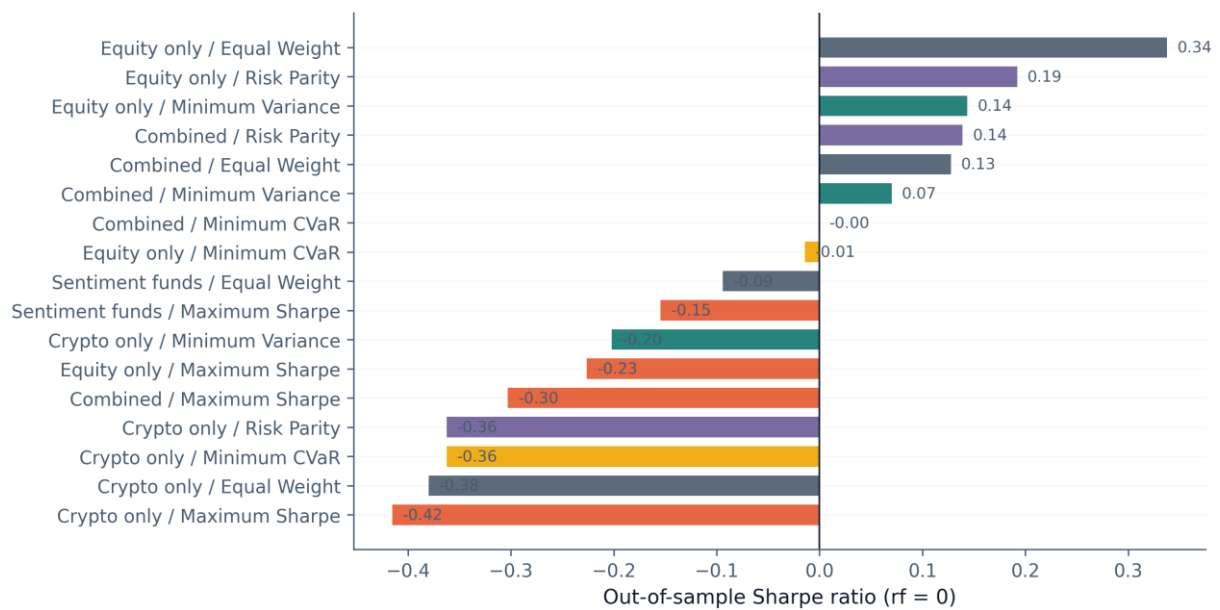


Figure 4. Out-of-sample Sharpe ratio ($rf = 0$) for all 17 funds, 2 Jan 2022 to 29 Dec 2023. Colours denote family.

2.4 A fund fact sheet

Every fund has a fact sheet that can be viewed in detail through the “Funds” tab of the Streamlit application. All fact sheets have the same fields of total and annualised return, annualised volatility, Sharpe, maximum drawdown, Calmar and growth of \$1. As an example, the combined Maximum Sharpe fact sheet reports a CAGR of -6.08%, volatility of 20.07%, Sharpe -0.303, maximum drawdown -29.28%, and growth of \$1 to \$0.88 net of fees. Table 3 lists its ten largest holdings and the 10% cap binds on six names (ABBV, AMGN, MRK, GE, XOM, OXY) and the remainder spreads across TMUS, SLB, SO and NUE. The full fact sheet for every fund with these same fields plus the current holdings from the last rebalance is reproduced in Appendix B (Tables B1-B17).

The best fund, which is the Equity Equal Weight portfolio, reports a total OOS return of 12.23%, CAGR +5.97%, annualised volatility 17.71%, Sharpe 0.338, maximum drawdown -20.32%, Calmar 0.29, and growth of \$1 to \$1.12 net of fees (Table A4). By construction it holds each of the 50 names at exactly 2.0% and is never rebalanced, so its turnover is zero. It beat every optimised equity fund in a period when getting the direction of the market wrong was expensive.

Table 3. Current holdings of combined Maximum Sharpe at the last monthly rebalance (29 December 2023).

Ticker	Weight
ABBV	10.0%
AMGN	10.0%
MRK	10.0%
GE	10.0%
XOM	10.0%
OXY	10.0%
TMUS	8.2%
SLB	7.0%
SO	7.0%
NUE	5.4%

3. The news sentiment index

3.1 From VADER to FinSent

To facilitate the sentiment analytics of AlphaStream, VADER, which is a rule-based lexicon scorer, has been extended by finance lexicon created for project known as FinSent. VADER on its own is poorly calibrated for financial headlines. This is also evident through the data as 48.1% of the 146,836 unique headlines score exactly zero, and many of those are false neutrals. To combat this, FinSent extends the analyzer in three steps. Firstly, roughly 80 finance words are added with hand-assigned valence and intensity (beat, beatable, downgrade, margin, beat-expectations, growth, overweight, underperform, and so on). Secondly, about 65 multi-word phrases ('beats earnings estimates', 'misses consensus', 'upgrade to overweight', 'surpasses expectations') are registered as VADER idiom phrases. Third, the headlines are scored exactly once and cached and the scores have been validated by eyeballing a random sample of 50 headlines to ensure the machine scoring is consistent with a human read. Figure 5 demonstrates the effect of this as the neutral fraction falls from 48.1% to 29.1%, a ~19% difference.

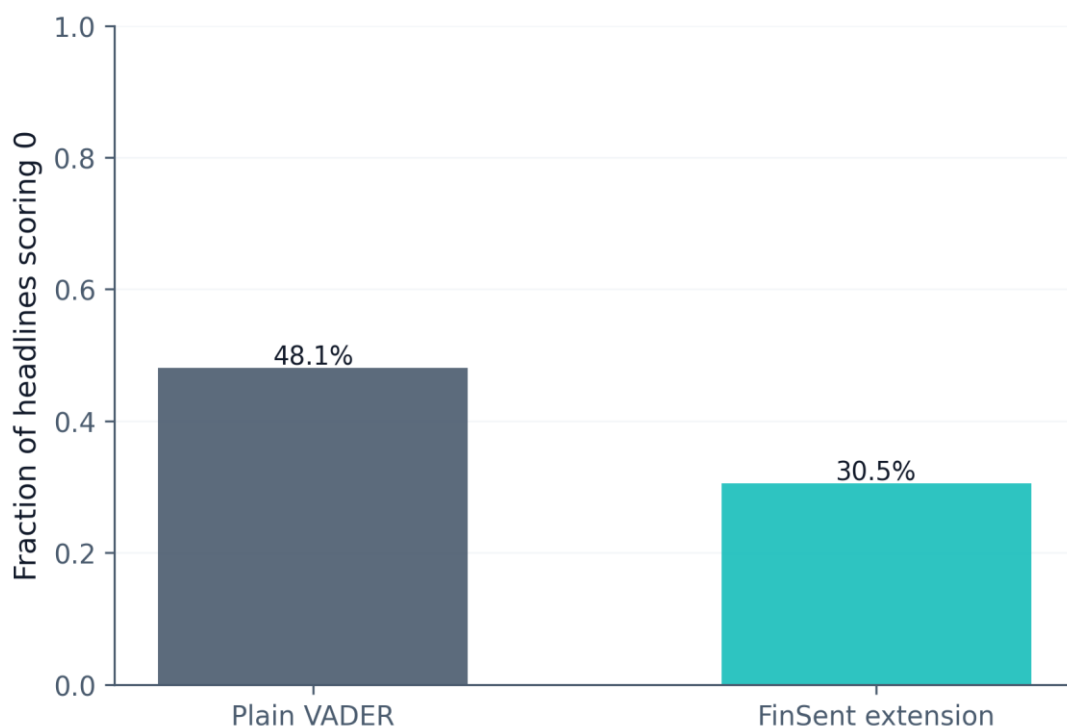


Figure 5. Fraction of unique headlines scoring exactly zero under plain VADER versus the FinSent extension, all 146,836 unique headlines, 2020-2023.

3.2 Building the sector index

Each headline is aligned to its equity trading day (same day if trading, otherwise the next trading day), scored and then averaged to a ticker day score. The sector index allocated equal weightage to the tickers in each of the ten sectors. In the construction of the design index two design choices were important. Firstly, no-news days are treated as neutral (0.0) rather than being dropped as dropping them would bias the sector average to days with news flow. Secondly, the signal is lagged one trading day before it can be used to make portfolio decisions, so a Saturday or Monday headline aligned to Monday is first usable for Tuesday's trade.

The pipeline scores unique titles and caches the score table to parquet so the build and the app never re-run the scorer. Ticker-day scores are the equal-weight average of that day's headline scores for the ticker; a ticker with several headlines in a day contributes its average, and not the sum. The sector index then averages the ticker-day scores within each sector. Casing and punctuation in the text is kept on

purpose as VADER treats all-caps and exclamation marks as intensity signals ('BEATS' is stronger than 'beats', 'great!' is stronger than 'great'), so stripping them would throw away signal. Coverage is an area in which the sectors differ however this is reported rather than hidden (Table 4).

Table 4. News coverage by sector: share of equity trading days with at least one headline, 2020-2023. Sparse sectors (Materials, Real Estate, Utilities) are the ones whose index entries most often rest on neutral 0.0.

Sector	% days with news	Sector	% days with news
Tech	99.8	Consumer	99.2
Financials	99.2	Healthcare	98.8
Energy	98.7	Comm	98.3
Industrials	98.1	Utilities	92.4
Real Estate	88.9	Materials	84.4

3.3 Reading the index

Figure 6 plots the market index (equal weight of sector indices) and the ten sector series. The market index is persistently positive with a mean of 0.100 and a standard deviation 0.040 over the 1,006 trading days. The market index is a sentiment level, not a return forecast: its mean of 0.10 reflects a press that mostly writes positive sentences, so using the raw level as a timing signal would have bought everything in every month. Rather, a cross-sectional spread is the signal as the gap between the highest and lowest sector on a given day carries the information that which names are loved, and which are feared. A fusion strategy should therefore concentrate on the regime of high dispersion.

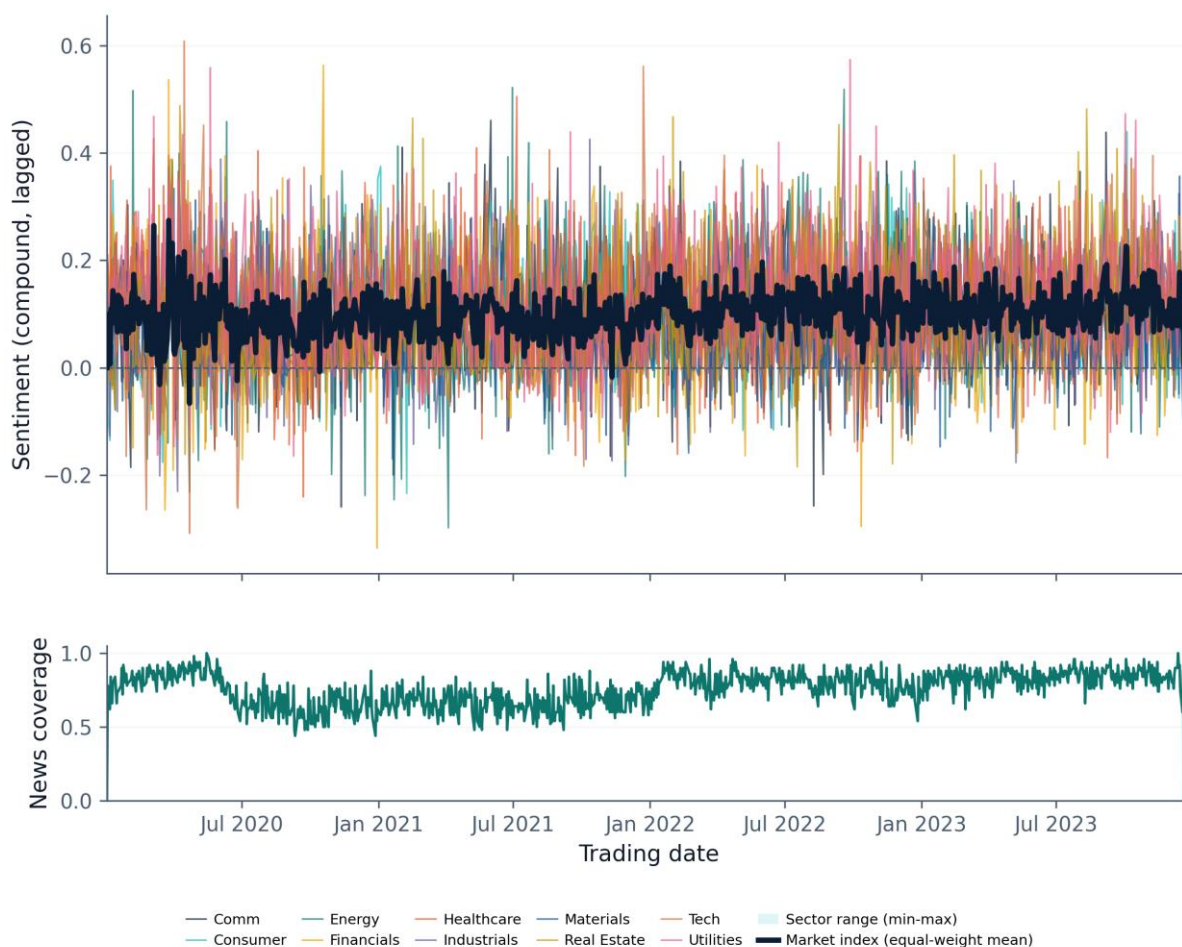


Figure 6. AlphaStream FinSent market index (thick line) and ten sector sentiment indices, 2020-2023, equal-weighted tickers, no-news days neutral, lagged one trading day.

4. Extensions and innovations

4.1 The innovation map

Innovation is a top priority for AlphaStream and Table 5 demonstrates these innovations are stated up front with their evidence rather than buried in the text. Every extension is implemented in the build script and reproduced in this report and the app. The extension with a negative result (the fusion, Section 4.4) and the loss-making crypto portfolios (Section 2) are kept and explained.

Table 5. The AlphaStream innovation portfolio: each extension, what it adds beyond the required baseline, and where its evidence appears in this report.

Extension	What it adds beyond the baseline	Evidence
FinSent lexicon	Custom sentiment tool: ~80 finance words, ~65 phrases fixing VADER's false neutral	Sec 3.1; Fig 5 (48.1% to 29.1% neutral)
17-fund shelf, five methods	Wider fund set than the two-method minimum; adds Minimum CVaR tail risk	Sec 1.2; Table 2
Turnover as a cost proxy	Average monthly one-way turnover, reported and interpreted as cost exposure	Sec 1.3; Table 2; Sec 6.3
Fusion designs	Two look-ahead-safe news uses: constraint-preserving tilt, long-short momentum fund	Sec 4.2-4.4; Table 6; Figs 7-8
AlphaStream design system	Original figure and app design system rather than the provided style	Sec 5.2; every figure and the app
Results-first thin app	App reads precomputed results only; never imports nltk or re-runs an optimiser	Sec 5.3
Reused Part A diagnostics	Spillover index and crypto-to-equity lead-lag used to interpret sentiment and fusion	Sec 1.1, 3.3; Tables A2-A3

4.2 Two fusion designs

The equity sentiment is integrated into the equity funds in two ways, both of which are look-ahead-safe. The first is through a tilt. At each monthly decision date, the equity Maximum-Sharpe weights are rewritten them by multiplying $(1 + a z)$, where z is the winsorised (within ± 2) cross-sectional z -score of each name's five-day mean of lagged FinSent and $a = 1.0$. Then the weights are renormalised and kept long-only, so the base fund's cap structure is preserved. With $a = 1.0$ the maximum tilt a name can receive is a tripling of its base weight (at $z = +2$). The second is a long-short momentum fund in which each month 50 equities are ranked by their lagged sentiment and hold the top ten and short the bottom ten. The base used for this strategy is the equal weight equity fund. The momentum fund is the aggressive variant of this strategy, and its results are a test of whether the FinSent ranking has any monthly persistence at all.

4.3 Before and after

Table 6 and Figures 7 and 8 compare each fusion fund with its base. The tilt improves on Maximum-Sharpe on every metric - Sharpe -0.226 to -0.154, CAGR -4.14% to -2.85%, maximum drawdown -25.74% to -24.40%. However, its CAGR is still slightly negative, because the base it tilts is the weakest equity fund. The momentum fund is the sharper negative result as it loses -0.72% a year with a Sharpe of -0.094 while its equal-weight base earns +5.97% with a Sharpe of 0.34. The only metric which outperforms is drawdown (-13.1% vs -20.3%).

Table 6. Fusion before-vs-after, net of fees, 2 Jan 2022 to 29 Dec 2023. The tilt is the Max-Sharpe fund tilted on FinSent; momentum is the long-short sentiment fund vs its equal-weight base.

Fund	CAGR	Volatility	Sharpe	Max DD	Turnover
Base (Max Sharpe)	-4.14%	18.33%	-0.226	-25.74%	0.34
Sentiment tilt	-2.85%	18.44%	-0.154	-24.40%	0.62
Base (Equal Weight)	+5.97%	17.71%	0.338	-20.32%	0.00
Sentiment momentum	-0.72%	7.61%	-0.094	-13.11%	1.63

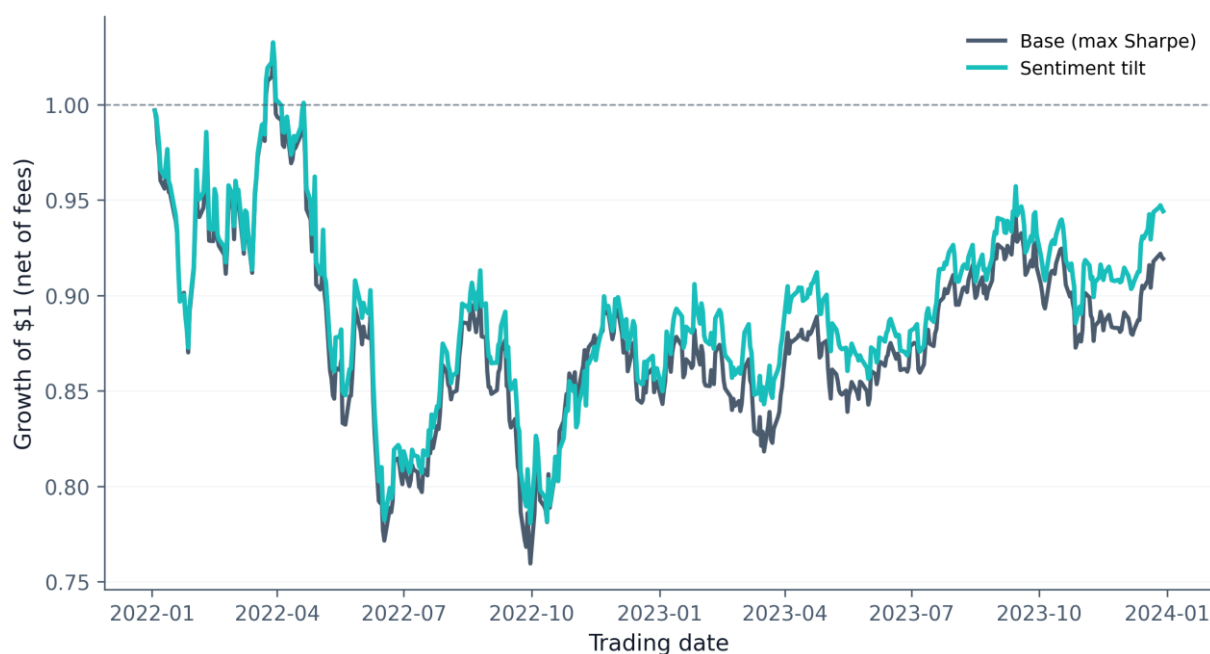


Figure 7. Growth of \$1 (net of fees): equity Maximum Sharpe versus the sentiment tilt, 2 Jan 2022 to 29 Dec 2023.

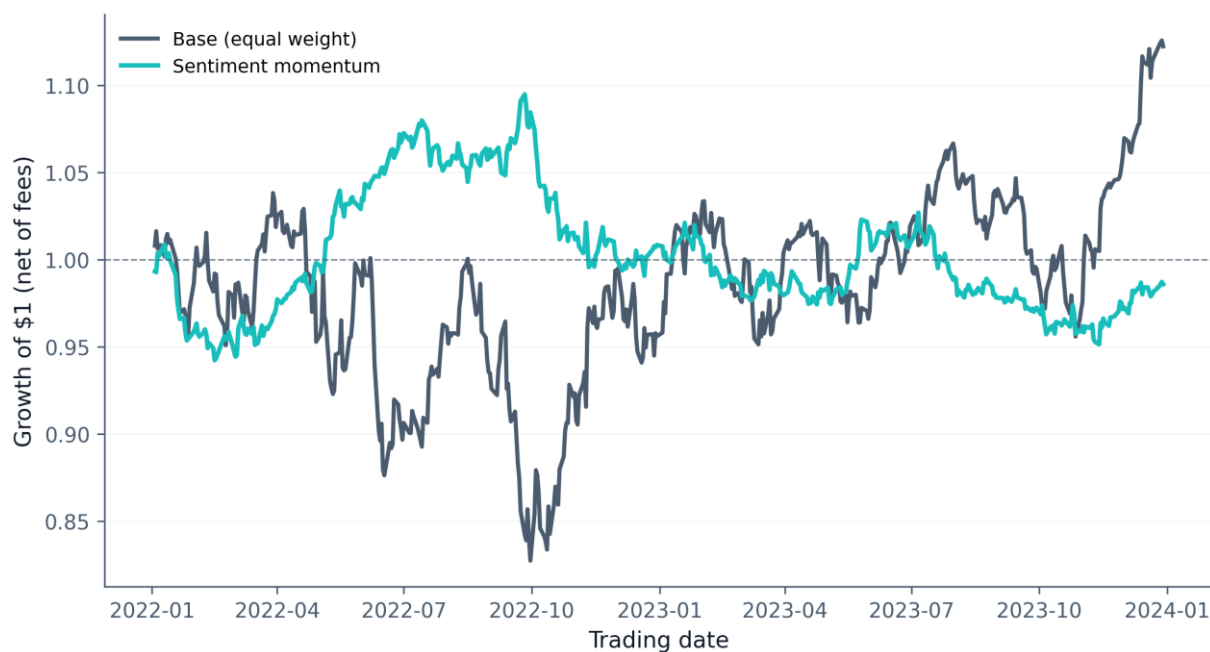


Figure 8. Growth of \$1 (net of fees): equity Equal Weight versus the sentiment-momentum long-short fund, 2 Jan 2022 to 29 Dec 2023.

4.4 What the negative result means

The fusion of sentiment with the equal weight equity portfolio produced negative results. This is likely as sentiment is a noisy proxy for tradeable information. Many headlines restate stale news, and the five-day average still carries that noise. Furthermore, monthly rebalancing is too slow for a daily signal as the tilt is forced to hold whatever the sentiment was at the decision date for the whole month, and most of the signal decays within days.

5. The app and the investor journey

5.1 Target user and journey

The target user for AlphaStream are individuals who want to systematically invest in equities and crypto. The journey for an individual using the Streamlit app comprises of first landing on the home tab and navigating through different tabs. Firstly, the funds tab displays the different types of funds and their corresponding OOS track record and fact sheet where there is also an option to compare all 17 side by side. The sentiment tab on the other hand allows the user to gauge the prevailing emotions of the market. The innovation tab provides an insight into the original and unique features of AlphaStream. Furthermore, the user can then also allocate capital across the four types of funds through the allocate tab and watch a blended fund-of-funds growth curve. Finally, a preview of the underlying raw data and integrity audit can be found on the data tab.

5.2 Design system

The app and the figures share the AlphaStream design system, so the product looks like one thing across surfaces. Figures are built by one figure module with a single theme and self-contained captions (title, sample period, units, source). The report embeds the same files the app serves, so there is no drift between the narrative and the product. The system is predicated upon discrete code rather than convention. `src/design.py` exports the palette and matplotlib style. Therefore, every chart in `results/figures` is produced by `src/figures.py` with it, so a change of brand colour is one edit.

5.3 Architecture and deployment

The app is deliberately thin. All ETL, scoring, optimisation, and backtesting run once in `scripts/run_part_b.py` and write CSVs and figures to results. The deployed app only reads those artifacts. It never imports nltk, never runs VADER, and never re-solves a portfolio, so it starts fast and stays within Streamlit Community Cloud's free-tier limits. The app entrypoint is `streamlit_app.py` at the repository root, and the repo is deployed from a public GitHub repository with the live app URL submitted alongside this report.

6. Critical reflection

6.1 What worked

The backtest engine is the strongest part of AlphaStream. It is genuinely out-of-sample and reuses one verified walk-forward loop across all 17 funds. Its assumptions (fees, rf, caps, calendar) are also explicit and consistent everywhere they appear. The FinSent lexicon is the clearest innovation and delivers promising results as seen by its verifiable improvement over VADER (48.1% to 29.1% false neutral).

A quieter success is integrity discipline. The ETL reports every anomaly it finds - 2,847 duplicate headlines, 10 stray 2024-01-01 crypto rows, 1,800 missing equity ticker-days that trace to 36 US holidays, 384 outlier returns kept in sample instead of silently repairing them. Additionally, working alongside and AI agent also increased the speed and efficiency of the project, however careful oversight was still necessary as the prompt log records substantive AI mistakes that were caught by verification rather than carried into the analysis (see ai/AI_NOTES.md).

6.2 What did not work

There were three predominant areas which did not work as initially anticipated. Firstly, the crypto allocation destroyed value as every crypto fund lost money OOS. The combined funds hold up to 20% crypto and as a result diversification across asset classes hampered key financial metrics rather than improving them in this specific window. Secondly, Maximum-Sharpe was the worst fund in every family as its mean-variance forecast is too unstable on daily data to be trusted monthly. Finally, the sentiment fusion did not add value beyond the equal equity portfolio which suggest that a daily signal rebalanced monthly cannot beat buy-and-hold equal weight in this sample.

6.3 Three recommendations

Three recommendations follow directly from the results.

Firstly, shrink the crypto allocation in the combined funds. The clearest failure in this window was the 20% crypto cap as every crypto fund lost between 11% and 24% a year, and the best combined fund (+2.67% CAGR, Equal Weight) earned less than half of the best equity fund (+5.97%). The combined fund needs either a much smaller crypto cap (for example 5-10%) or a volatility-targeting layer that trims the allocation as crypto volatility spikes. Both are a parameter change in the existing optimiser, and the walk-forward backtest can re-rank the funds under the new setting in minutes.

Secondly, the Maximum Sharpe should not be used when investing. It was the worst fund in every family and also proved to be one of the most expensive to run (0.31-0.34 monthly turnover). Its mean-variance forecast is too noisy on a 504-day sample to be trusted monthly. Therefore, if a combined fund is sought by an investor, then the app should suggest the two funds that made money which were Equal Weight, Risk Parity or Minimum Variance.

Thirdly, use sentiment only where the news is real and fresh. Section 4.4 traced the fusion failure to the signal decay within days while the tilt holds it for a month. The practical fix is to match the holding period to the signal's decay. Each variant should be tested with the same before-vs-after table as Table 6 and should also be graphed as seen in Figure 7 and Figure 8 for effective visualization, and a version that still underperforms should be reported as a negative result rather than tuned away.

Appendix A. Additional exhibits

Required and supporting exhibits referenced from the main text are collected here. All use the same OOS period (2 Jan 2022 to 29 Dec 2023) and the same net-of-fee convention as the main tables and figures.

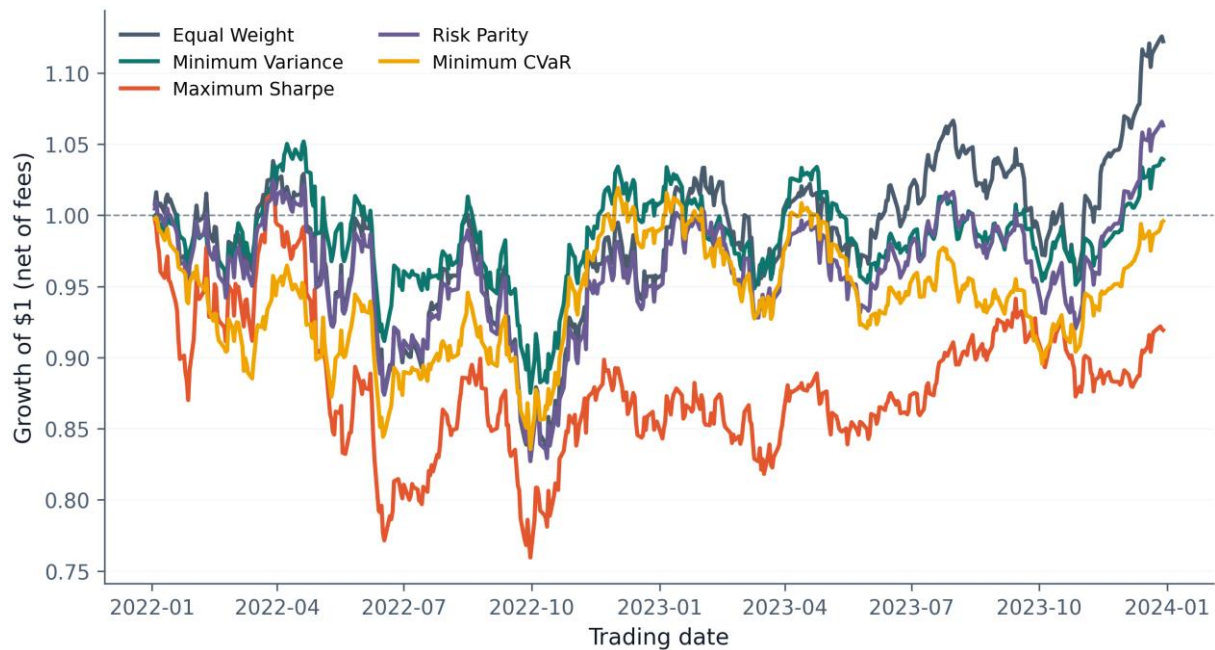


Figure A1. Growth of \$1 (net of fees) for the five equity funds, 2 Jan 2022 to 29 Dec 2023.

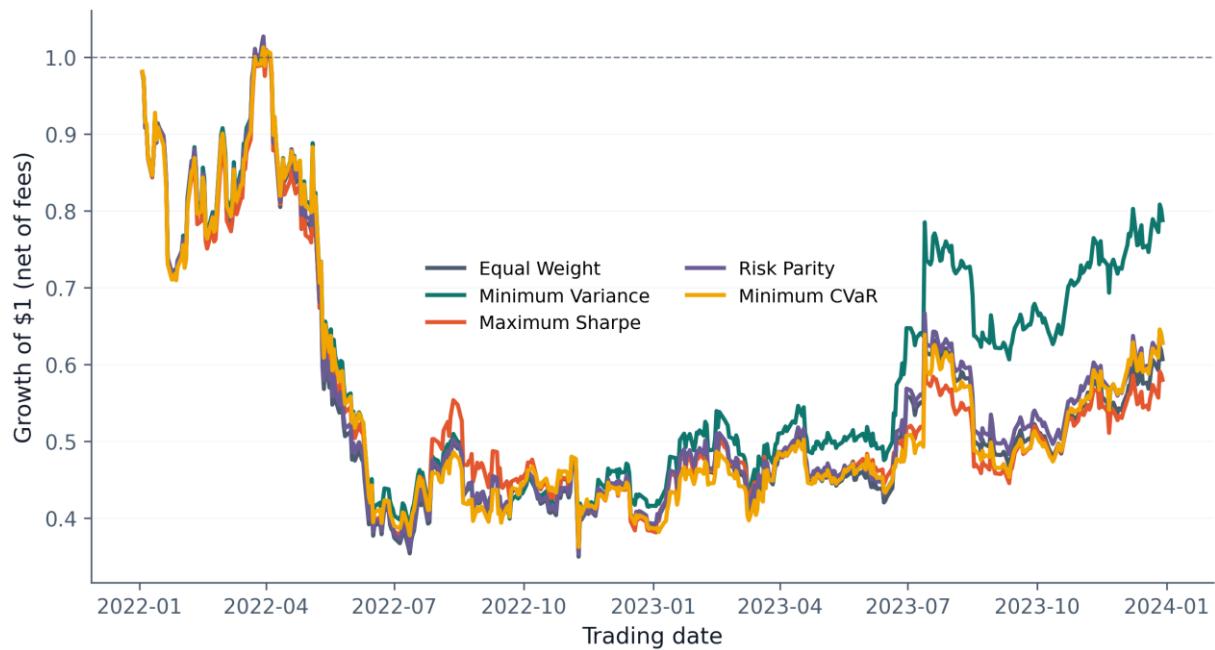


Figure A2. Growth of \$1 (net of fees) for the five crypto funds, 2 Jan 2022 to 29 Dec 2023. Note the scale: every crypto fund loses more than half its value.

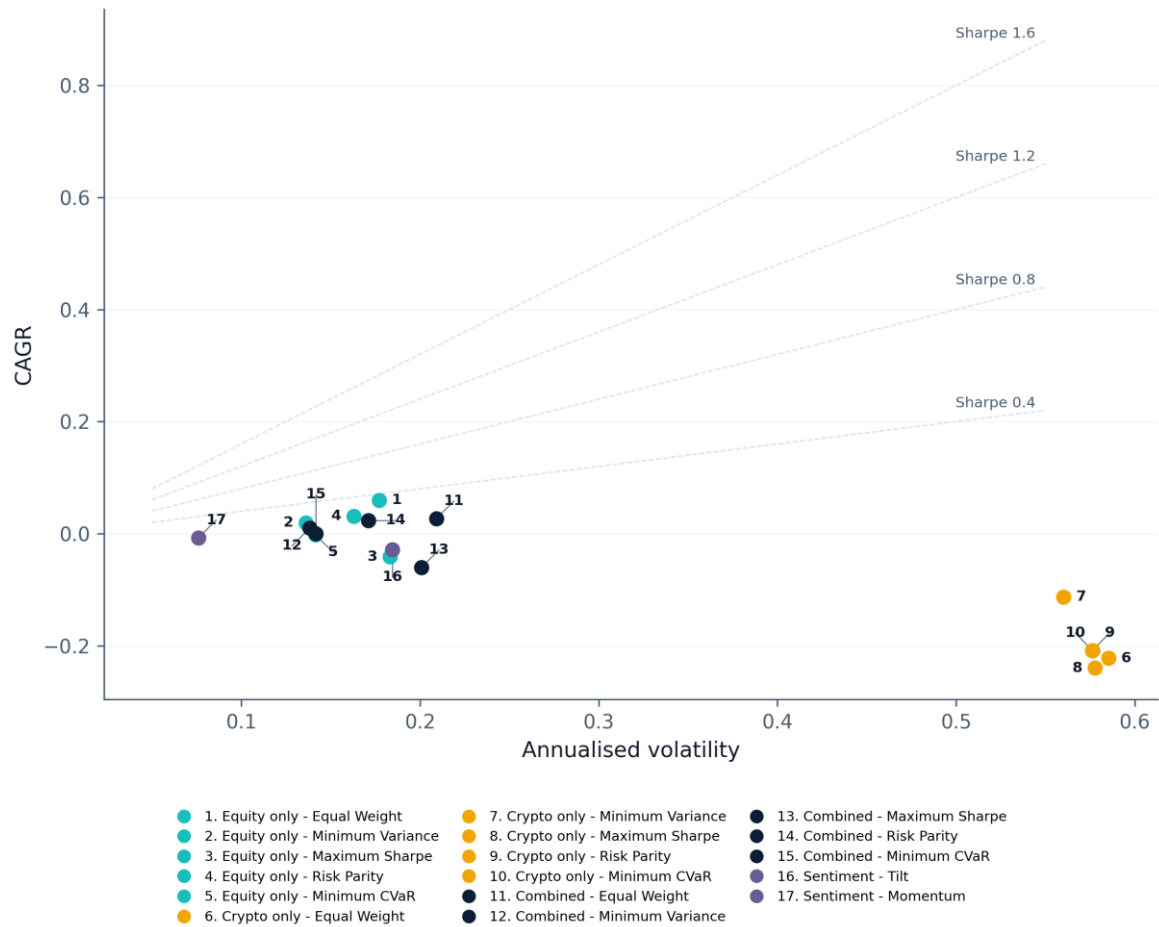


Figure A3. Annualised return vs annualised volatility for all 17 funds, with iso-Sharpe lines, 2 Jan 2022 to 29 Dec 2023. Each point is numbered and keyed in the legend; point colour is the asset family (equity cyan, crypto amber, combined navy, sentiment violet).

Table A1. Station 1 integrity audit reproduced for Part B (from results/tables/integrity_report.csv).

Dataset	Check	Count
Equity prices	Rows loaded	50,300
Equity prices	Duplicate ticker-date pairs removed	0
Equity prices	Missing trading dates	1,800
Equity prices	Outlier returns flagged (kept)	384
Crypto prices	Rows loaded	14,620
Crypto prices	Stray 2024-01-01 rows removed	10
Crypto prices	Rows after cleaning	14,610
Crypto prices	Duplicate ticker-date pairs removed	0
Crypto prices	Missing calendar dates	0
News headlines	Rows loaded	149,683
News headlines	Exact duplicate rows removed	2,847
News headlines	Rows after deduplication	146,836

Table A2. Crypto-to-equity lead-lag diagnostic: the ten equities with the largest average absolute correlation with a lagged crypto factor. Peak correlations sit at lag 0 (same-day co-movement); lagged structure is weak.

Ticker	Max + corr	Lag	Max - corr	Lag	Avg corr
TMUS	0.318	0	-0.117	1	0.083
ADBE	0.351	0	-0.143	1	0.077
CMCSA	0.286	0	-0.115	1	0.076
DIS	0.355	0	-0.118	-1	0.068
V	0.309	0	-0.105	-1	0.067

O	0.191	0	-0.105	-1	0.067
INTC	0.289	0	-0.120	1	0.065
QCOM	0.330	0	-0.106	1	0.064
DUK	0.180	0	-0.077	1	0.063
SBUX	0.298	0	-0.081	1	0.062

Table A3. Cross-sector news-flow spillover index (Part A feature reused), 2020-2023.

Statistic	Value
Observations	992
Mean	0.172
Std dev	0.080
Maximum	0.710
Minimum	0.038

Table A4. Fact sheet for the best-performing fund, equity Equal Weight, net of fees, 2 Jan 2022 to 29 Dec 2023.

Metric	Value
Total return (ITD)	12.23%
CAGR	5.97%
Annualised volatility	17.71%
Sharpe ratio	0.337
Max drawdown	-20.32%
Calmar	0.29
Growth of \$1 (net of fees)	1.11

Appendix B. Fund fact sheets

Table B1. Fact sheet for Equity Equal Weight, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	12.23%
CAGR	5.97%
Annualised volatility	17.71%
Sharpe ratio	0.337
Max drawdown	-20.32%
Calmar	0.29
Growth of \$1 (net of fees)	1.11
Current top five holdings	ABBV 2.0%; ABT 2.0%; ADBE 2.0%; AEP 2.0%; AMD 2.0%

Table B2. Fact sheet for Equity Minimum Variance, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	3.93%
CAGR	1.96%
Annualised volatility	13.62%
Sharpe ratio	0.144
Max drawdown	-16.80%
Calmar	0.12
Growth of \$1 (net of fees)	1.03
Current top five holdings	WMT 10.0%; MRK 10.0%; ABBV 9.4%; KO 9.2%; AMGN 8.9%

Table B3. Fact sheet for Equity Maximum Sharpe, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-8.07%
CAGR	-4.14%
Annualised volatility	18.33%
Sharpe ratio	-0.226
Max drawdown	-25.74%
Calmar	-0.16
Growth of \$1 (net of fees)	0.91
Current top five holdings	ABBV 10.0%; AMGN 10.0%; GE 10.0%; MRK 10.0%; OXY 10.0%

Table B4. Fact sheet for Equity Risk Parity, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	6.31%
CAGR	3.13%
Annualised volatility	16.29%
Sharpe ratio	0.192
Max drawdown	-19.25%
Calmar	0.16
Growth of \$1 (net of fees)	1.05
Current top five holdings	MRK 4.3%; ABBV 3.5%; WMT 3.4%; AMGN 3.2%; KO 3.0%

Table B5. Fact sheet for Equity Minimum CVaR, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-0.41%
CAGR	-0.20%
Annualised volatility	14.15%
Sharpe ratio	-0.014
Max drawdown	-16.31%

Calmar	-0.01
Growth of \$1 (net of fees)	0.99
Current top five holdings	KO 10.0%; TMUS 10.0%; V 10.0%; WMT 10.0%; MRK 10.0%

Table B6. Fact sheet for Crypto Equal Weight, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-39.34%
CAGR	-22.24%
Annualised volatility	58.53%
Sharpe ratio	-0.380
Max drawdown	-65.87%
Calmar	-0.34
Growth of \$1 (net of fees)	0.60
Current top five holdings	ADA-USD 10.0%; BCH-USD 10.0%; BTC-USD 10.0%; EOS-USD 10.0%; ETC-USD 10.0%

Table B7. Fact sheet for Crypto Minimum Variance, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-21.25%
CAGR	-11.32%
Annualised volatility	56.02%
Sharpe ratio	-0.202
Max drawdown	-64.04%
Calmar	-0.18
Growth of \$1 (net of fees)	0.78
Current top five holdings	BTC-USD 20.0%; TRX-USD 20.0%; ETH-USD 17.9%; XLM-USD 10.2%; BCH-USD 9.1%

Table B8. Fact sheet for Crypto Maximum Sharpe, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-42.04%
CAGR	-23.99%
Annualised volatility	57.77%
Sharpe ratio	-0.415
Max drawdown	-64.90%
Calmar	-0.37
Growth of \$1 (net of fees)	0.58
Current top five holdings	BTC-USD 20.0%; ETC-USD 20.0%; BCH-USD 20.0%; XRP-USD 20.0%; TRX-USD 20.0%

Table B9. Fact sheet for Crypto Risk Parity, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-37.21%
CAGR	-20.87%
Annualised volatility	57.62%
Sharpe ratio	-0.362
Max drawdown	-65.63%
Calmar	-0.32
Growth of \$1 (net of fees)	0.62
Current top five holdings	TRX-USD 14.2%; BTC-USD 12.4%; ETH-USD 9.9%; BCH-USD 9.5%; LTC-USD 9.4%

Table B10. Fact sheet for Crypto Minimum CVaR, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-37.23%
CAGR	-20.89%
Annualised volatility	57.65%
Sharpe ratio	-0.362
Max drawdown	-64.25%
Calmar	-0.33
Growth of \$1 (net of fees)	0.62
Current top five holdings	BCH-USD 20.0%; BTC-USD 20.0%; ETH-USD 20.0%; TRX-USD 20.0%; LTC-USD 11.0%

Table B11. Fact sheet for Combined Equal Weight, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	5.37%
CAGR	2.67%
Annualised volatility	20.91%
Sharpe ratio	0.128
Max drawdown	-26.94%
Calmar	0.10
Growth of \$1 (net of fees)	1.05
Current top five holdings	ABBV 1.7%; ABT 1.7%; ADBE 1.7%; AEP 1.7%; AMD 1.7%

Table B12. Fact sheet for Combined Minimum Variance, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	1.93%
CAGR	0.97%
Annualised volatility	13.83%
Sharpe ratio	0.070
Max drawdown	-17.52%
Calmar	0.06
Growth of \$1 (net of fees)	1.01
Current top five holdings	WMT 10.0%; MRK 10.0%; ABBV 9.0%; AMGN 8.1%; KO 8.0%

Table B13. Fact sheet for Combined Maximum Sharpe, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-11.72%
CAGR	-6.08%
Annualised volatility	20.07%
Sharpe ratio	-0.303
Max drawdown	-29.28%
Calmar	-0.21
Growth of \$1 (net of fees)	0.88
Current top five holdings	ABBV 10.0%; AMGN 10.0%; MRK 10.0%; GE 10.0%; XOM 10.0%

Table B14. Fact sheet for Combined Risk Parity, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	4.78%
CAGR	2.38%
Annualised volatility	17.10%
Sharpe ratio	0.139
Max drawdown	-21.80%

Calmar	0.11
Growth of \$1 (net of fees)	1.04
Current top five holdings	MRK 4.0%; ABBV 3.3%; WMT 3.2%; AMGN 3.0%; KO 2.8%

Table B15. Fact sheet for Combined Minimum CVaR, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-0.02%
CAGR	-0.01%
Annualised volatility	14.16%
Sharpe ratio	-0.001
Max drawdown	-16.31%
Calmar	-0.00
Growth of \$1 (net of fees)	0.99
Current top five holdings	KO 10.0%; TMUS 10.0%; V 10.0%; WMT 10.0%; MRK 10.0%

Table B16. Fact sheet for Equity Sentiment Tilt, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-5.58%
CAGR	-2.85%
Annualised volatility	18.44%
Sharpe ratio	-0.154
Max drawdown	-24.40%
Calmar	-0.12
Growth of \$1 (net of fees)	0.94
Current top five holdings	ABBV 15.4%; AMGN 11.5%; SO 11.5%; XOM 9.7%; NUE 9.5%

Table B17. Fact sheet for Equity Sentiment Momentum, net of fees, 2 Jan 2022 to 29 Dec 2023. Holdings as at 29 December 2023.

Metric	Value
Total return (ITD)	-1.42%
CAGR	-0.72%
Annualised volatility	7.61%
Sharpe ratio	-0.094
Max drawdown	-13.11%
Calmar	-0.05
Growth of \$1 (net of fees)	0.98
Current long top five	ABBV, AMD, GILD, D, O
Current short top five	ADBE, AEP, DD, MS, PSA